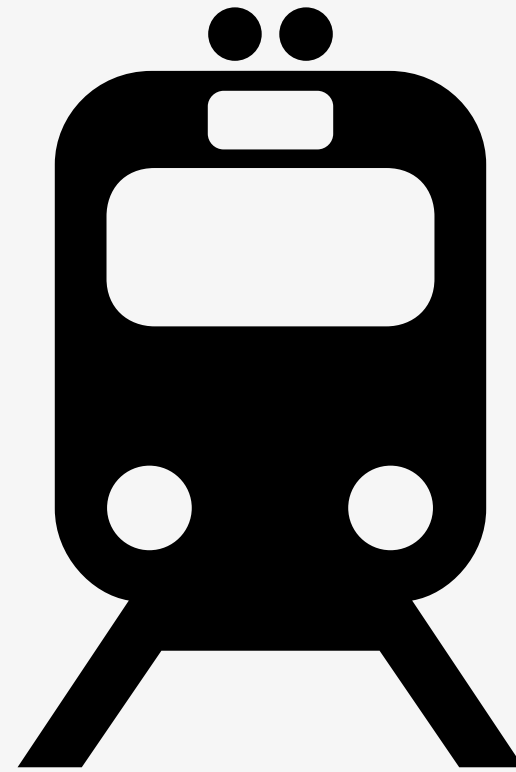


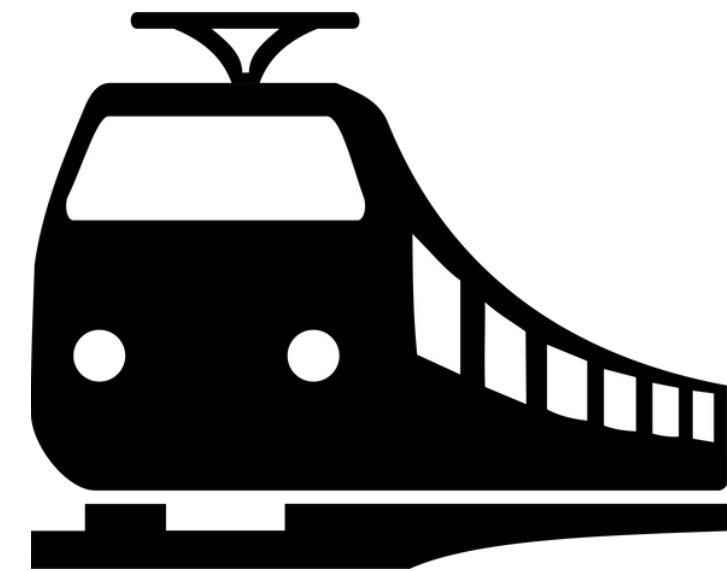
SUSTAINABLE CITIES



BUILDING AN INTERMODAL
TRANSPORTATION SYSTEM
BETWEEN BUSES AND METRO

AWADIS SHIKOYAN

TABLE OF CONTENTS



01 Introduction

03 Background and Justification

02 Problem and Statement

04 Finding and Analysis

05 Recommendations

INTRODUCTION

This research paper examines the potential for implementing an intermodal transportation system between buses and metro in Yerevan, Armenia. Using a combination of quantitative and qualitative methods, this paper explores existing public transportation infrastructure, intermodal transportation trends, and the public's preferences for transportation options. The following methods were used to enhance the study...

Data Analysis and Collection - by analyzing the dataset from gg taxi in the year 2016 and examining what the trends and frequencies of taxi orders are from the metro stations

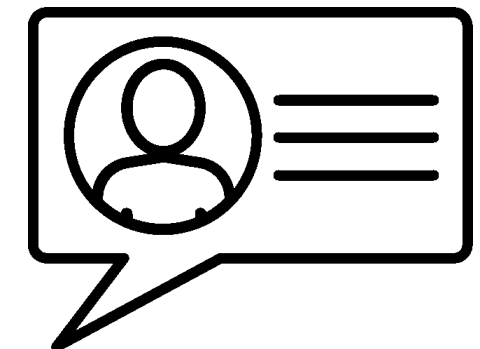


Manual Data collection- by manually collecting the waiting time data and bus/metro line connections to strategic parts of the city



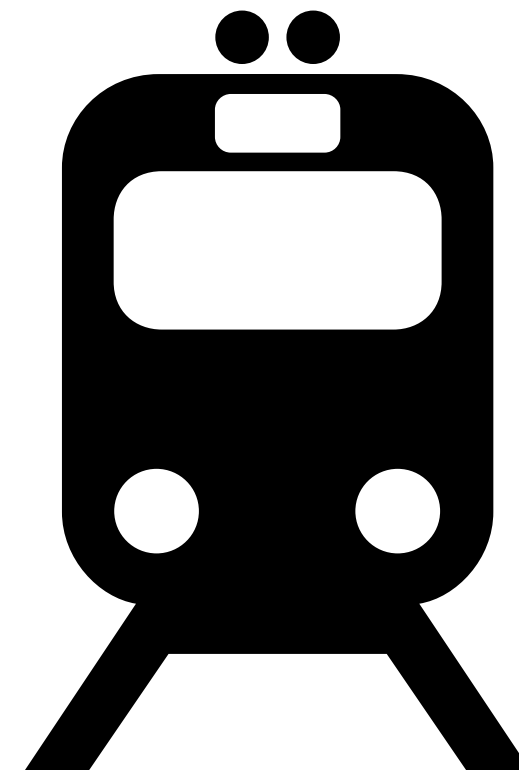
Live field observations

Analysis - trying to come up with appropriate recommendations based on the data



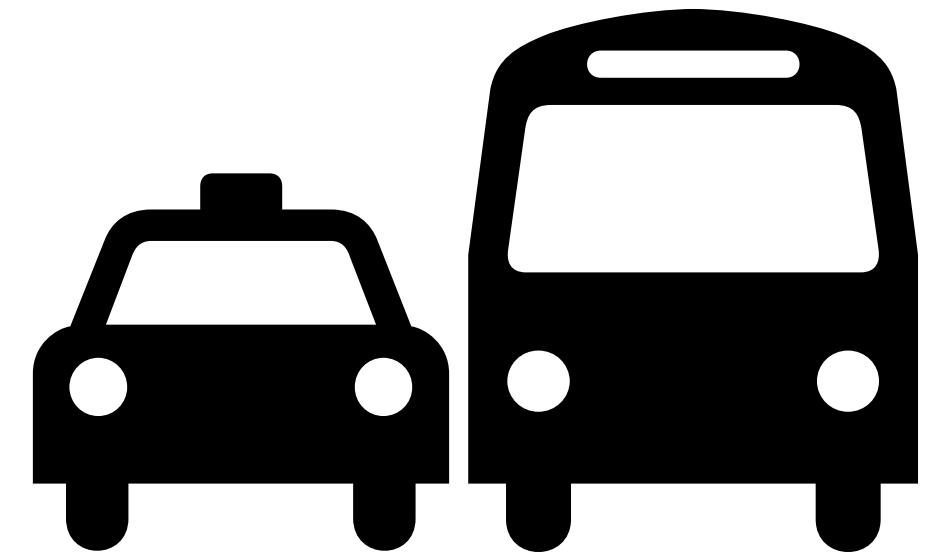
PROBLEM STATEMENT

Yerevan is in need of an effective and efficient means of integrating intermodal transportation between metros and buses in order to reduce traffic congestion and improve access to public transportation. The current transportation infrastructure is inadequate and unreliable, resulting in poor service quality, long wait times. There is a need to explore and implement strategies to improve the integration of intermodal transportation between metros and buses in Yerevan, such as improving the accessibility of public transportation, increasing the frequency and reliability of service.



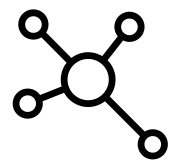
BACKGROUND AND JUSTIFICATION

Intermodal transportation is key to any successful urban transportation network. Combining the use of metro and bus services offers commuters a more convenient and efficient way to get around

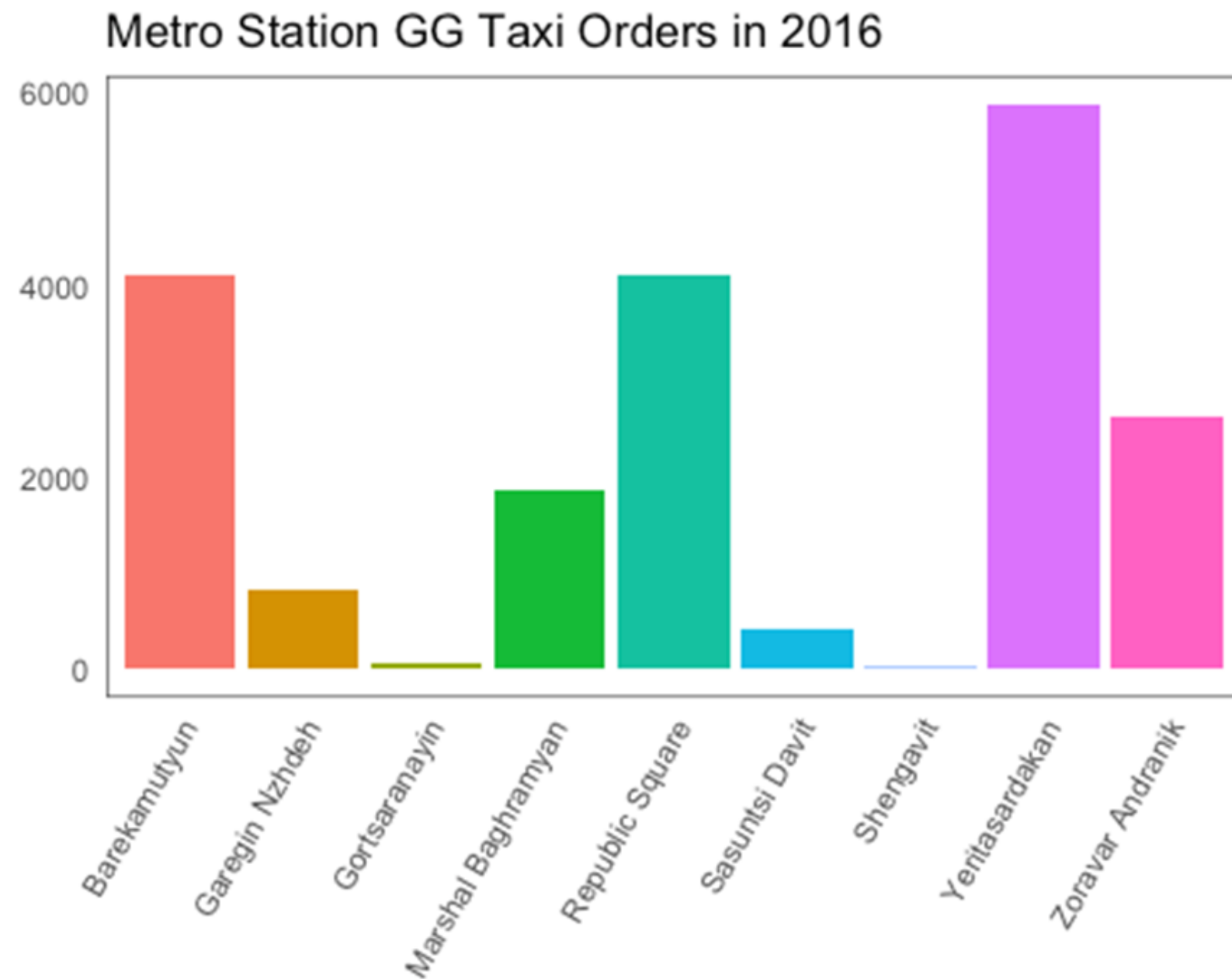


FACTORS OF SUCCESSFUL INTEGRATION

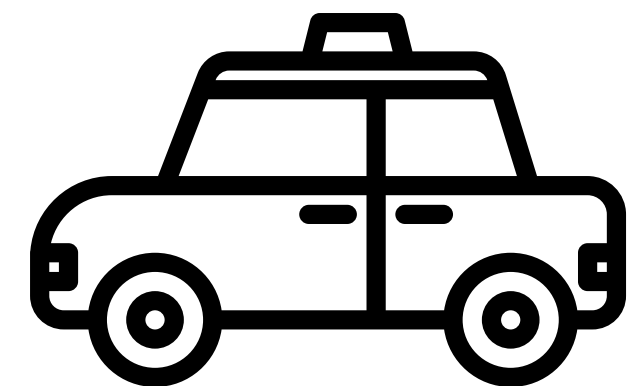
- Connectivity: Metros and buses should be conveniently connected to one another with reliable transfer points. Cities should also consider providing access to both modes of transportation from the same station or stop.
- Service Quality: Both metro and bus services should have high-quality, frequent, and reliable service in order to provide commuters with the best possible experience.
- Cost: The cost of using both services should be competitive with other forms of transportation.
- Planning: Cities should create a detailed plan that outlines the integration of metro and bus services. This plan should include the routes, frequencies, and fares of both services, as well as any special services or discounts that may be offered.



FINDINGS AND ANALYSIS

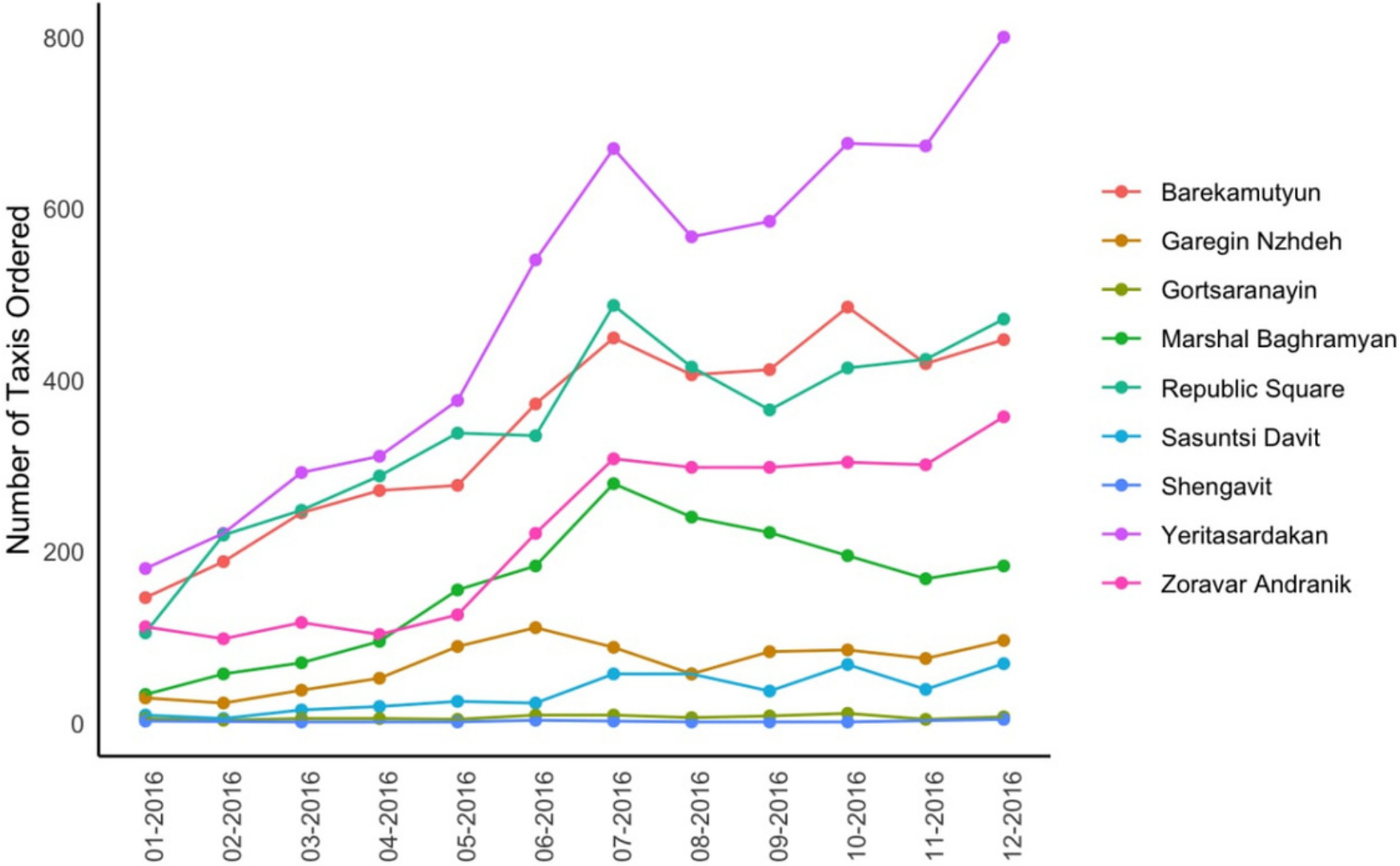


- Very high figures for metro stations that are concentrated more to the center of the city.
- Considering GG taxi is the second most user taxi ordering service in Yerevan, this figure in actuality is expectedly much higher.
- Do the stations provide enough coverage to reach all parts of the city?
- Do commuters have walkable and short pathways from when they leave the metro to a bus stop?



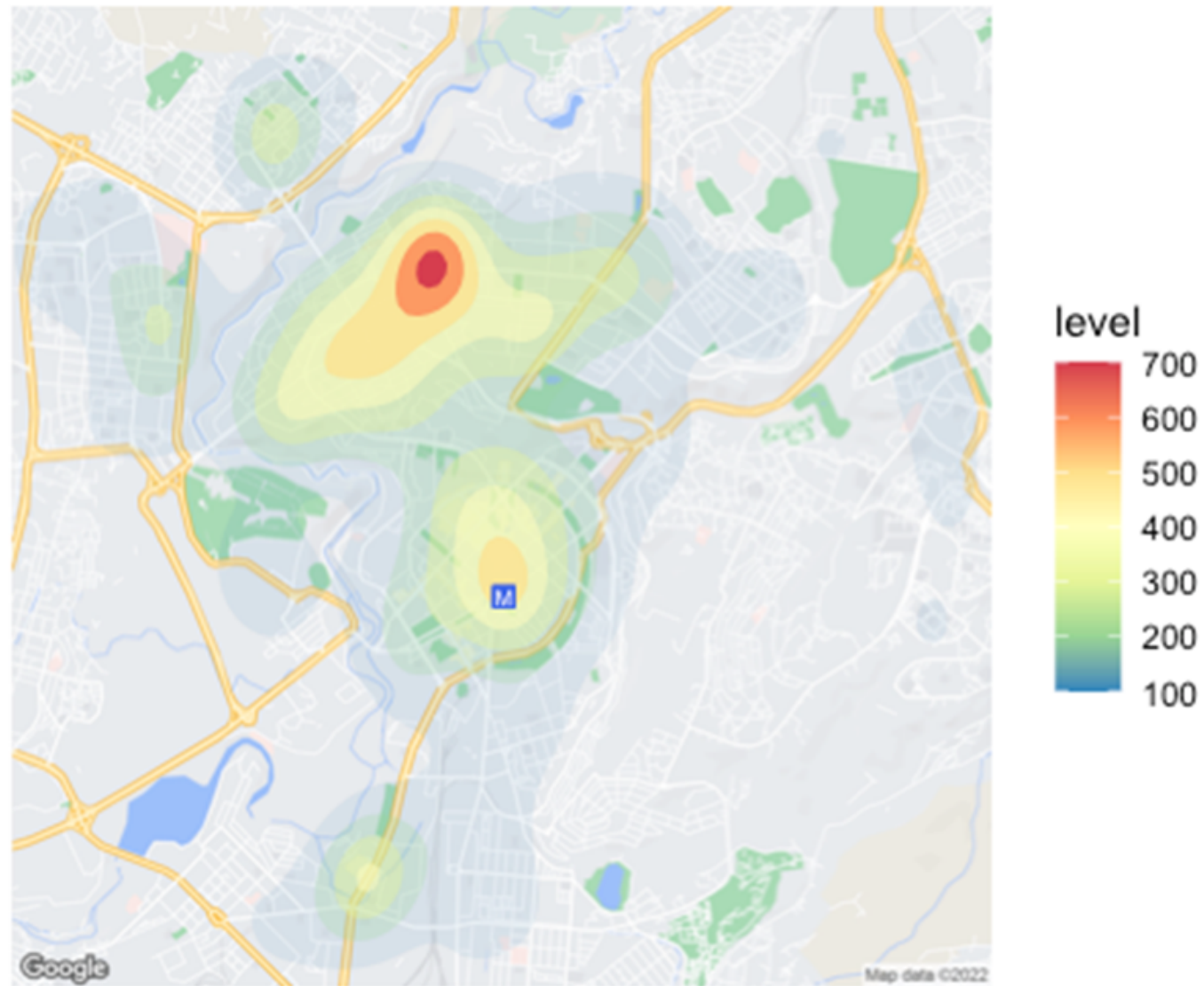
FINDINGS AND ANALYSIS

From 1st figure, we can predict would see most of the higher observations on the stations that are close to the center. We don't have any outliers or any data that would show us their inconsistencies in data. Thus we can go ahead and look at the metro stations that have the highest taxi order, counts.

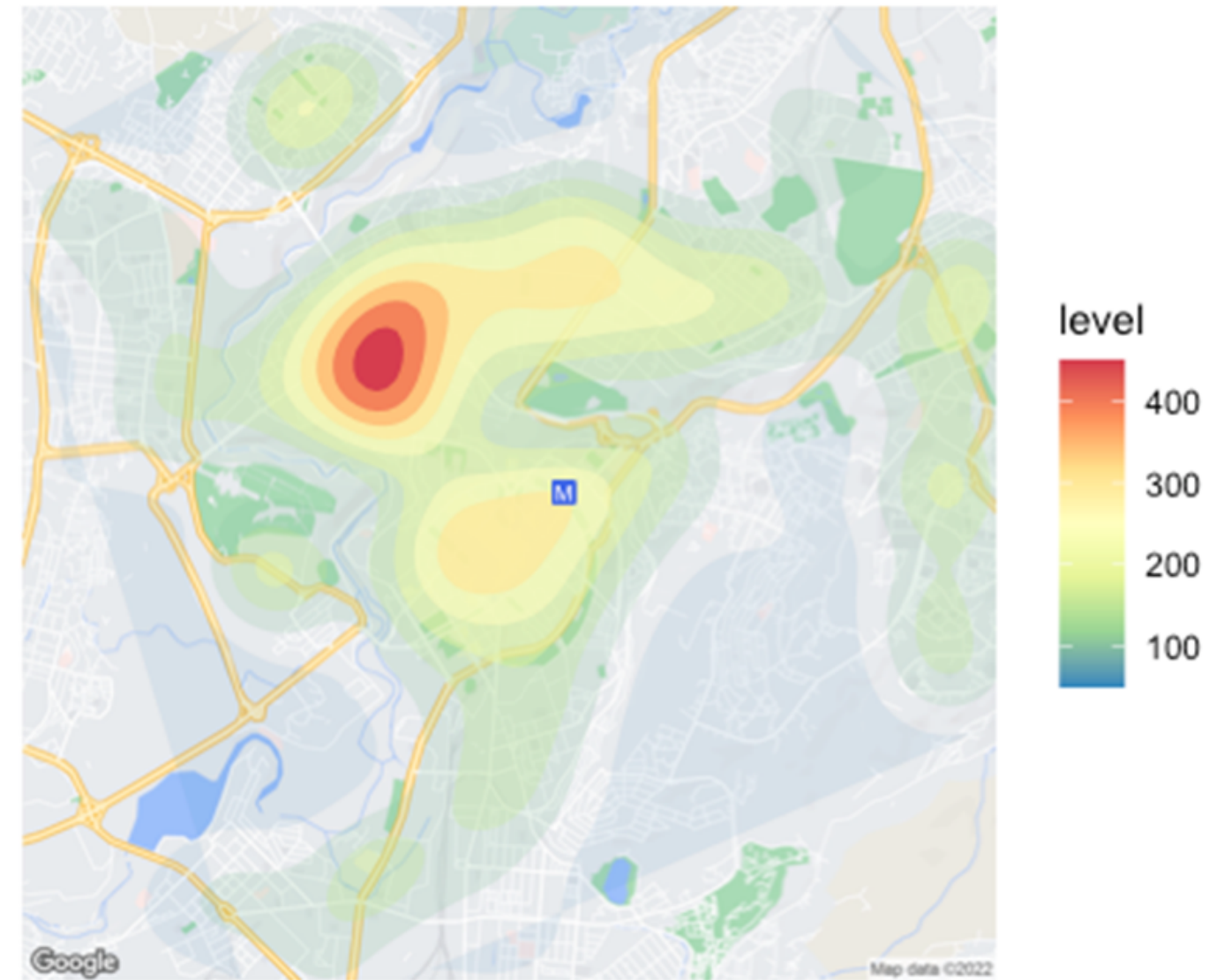


FINDINGS AND ANALYSIS

Taxi Order Destinations from Republic Square

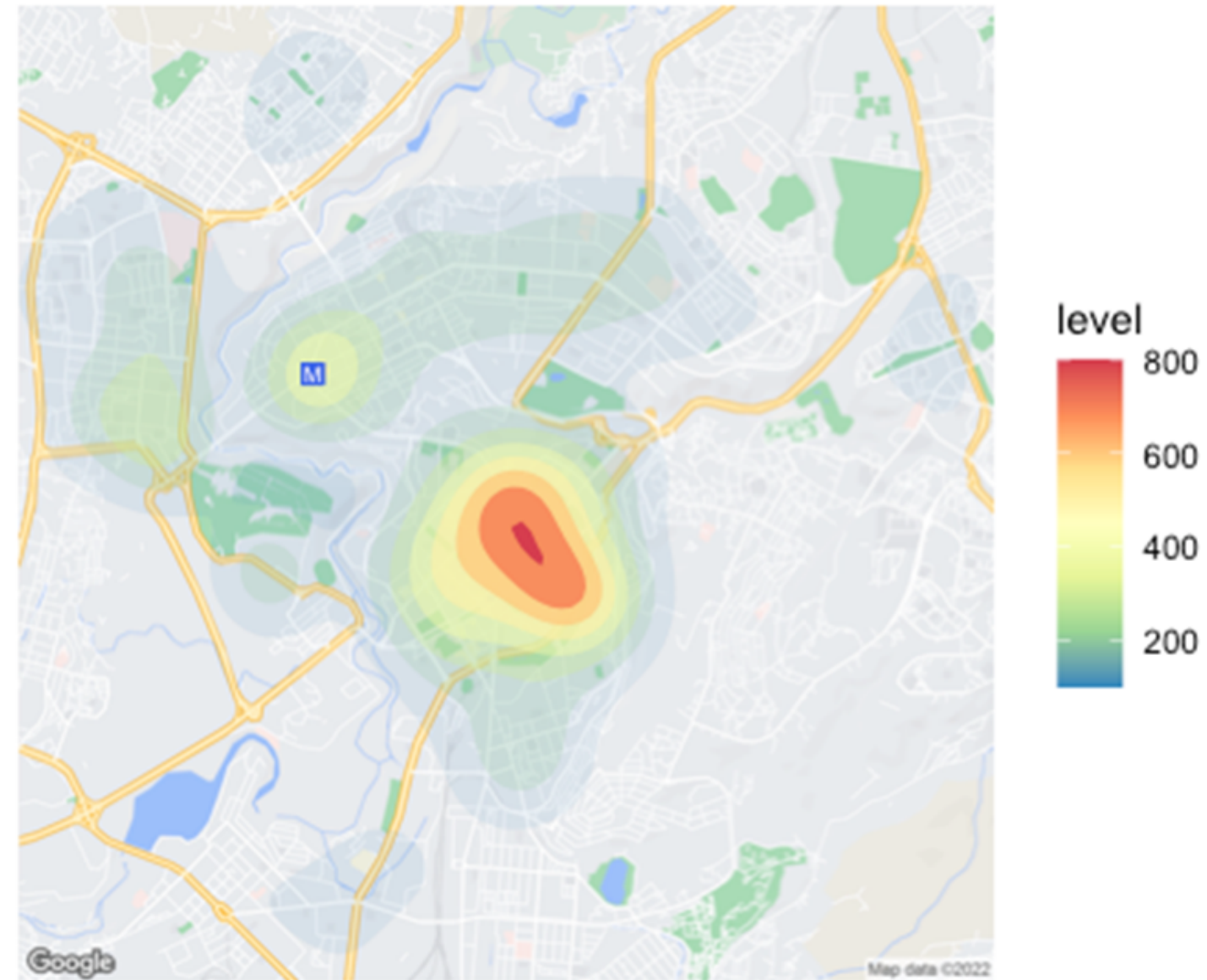


Taxi Order Destinations from Yeritasardakan



FINDINGS AND ANALYSIS

Taxi Order Destinations from Barekamutyun



FINDINGS AND ANALYSIS

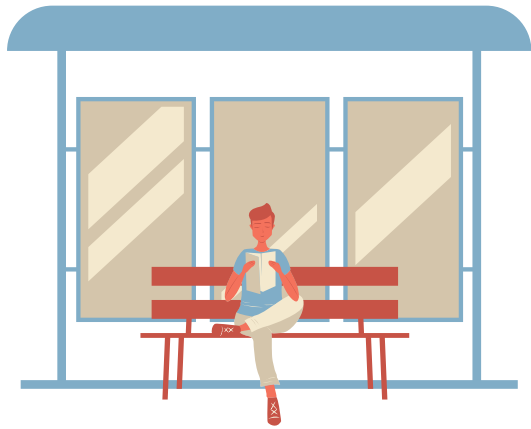
Republic Square Metro Station			
Region	Commute to the bus stop (in mins)	Available buses to get to the region	Commute to the region with a bus (in mins)
Arabkir	7-10	50, 10, 14, 46, (minibuses: 19, 28)	15-20
Kentron (Mastots Tumanyan)	8-10mins	16, 29, 201, 44 (13, 24, 33, 38)	10
Davtashen	5-7	74, 45	20-25
Ajapniak	15-18	(minibus 70, 26) and 12, 39, 67 (minibus 114, 22)	25*
Nor Nork	7-12	22, 42, 53, 63	35-40
Erebuni	9-12	41, 46, 97 (minibus 50)	19-22
* - you would need to change to two different buses			

Republic Yeritasardakan Station			
Region	Commute to the bus stop (in mins)	Available buses to get to the region	Commute to the region with a bus (in mins)
Arabkir	10	13, 47, 58, 97 (minibuses: 24)	15-17
Kentron (Mastots Tumanyan)	6-9	5, 26, 35, 54 (33, 77)	5-7
Davtashen	6-12	74, 45	15-20
Ajapniak	10	(minibus 70) and 12, 39, 67 (minibus 114, 22)	25*
Nor Nork	8	8, 22, 26, 35, 42	25
Erebuni	5-6	41, 46, 74 (minibus 50)	25-30
* - you would need to change to two different buses			

Marshal Baghramyan Metro Station			
Region	Commute to the bus stop (in mins)	Available buses to get to the region	Commute to the region with a bus (in mins)
Arabkir	4-5	13, 47, 11, 58 (minibuses: 4, 39)	15
Davtashen	4	45, 74	17
3rd Mas	5-7	22, 42, 53, 63	35
Ajapniak	7	74, 45 and (minibus 35, 57)	30*
Nor Nork	6	8, 35	30
Erebuni	9-12	41, 46, 97 (minibus 50)	19-22
* - you would need to change to two different buses			

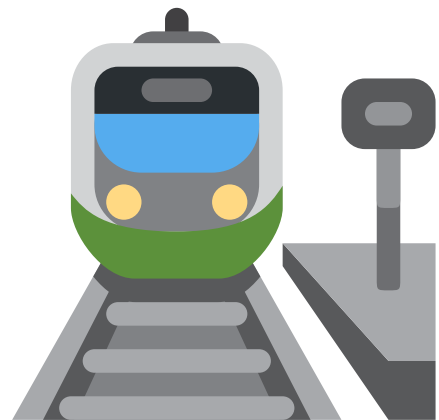
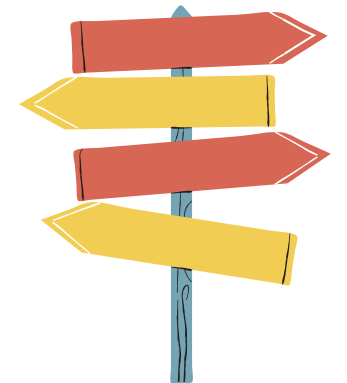
Barekamutyun Metro Station			
Region	Commute to the bus stop (in mins)	Available buses to get to the region	Commute to the region with a bus (in mins)
Arabkir	6-10	11, 13, 28, 3, 31, 58 (minibuses: 114, 12, 24)	10-12
Kentron (Mastots Tumanyan)	3-5	13, 47, 58 (minibus 4, 24)	13
Davtashen	4	74, 45	12
3rd Mas	5-7	22, 42, 53, 63	35
Ajapniak	16	3, 30, 35 (minibus 114, 34)	13
Nor Nork	7-11	3, 8, 35	40
Erebuni	5-12	41, 46, 74, 18 (minibus 68, 38)	25
* - you would need to change to two different buses			

RECOMMENDATIONS



Improving public transportation infrastructure of bus stops to be closer to metro stations

Increasing public education regarding available transportation options



Creating incentives for intermodal transportation.

Implemented more dockless bike and scooter-sharing services for short-distance travel in the city

